

RAHUL KUMAR

Software Engineer | Chandigarh, India

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Summary

Software Engineer experienced shipping production systems in **TypeScript, Go, and Rust**. Skilled at crafting polished, accessible frontend components in React/Next.js, building reliable services in **Rust**, and engineering high-performance APIs and data pipelines end-to-end, with a focus on **caching, security, and low-latency query scaling**.

Technical Skills

Languages: TypeScript, JavaScript, Go, Rust, SQL

Frontend: Next.js, React.js, Redux Toolkit, Tailwind CSS, Shadcn UI

Backend: Node.js, Express.js, Fastify, Echo, Fiber, gRPC, WebSockets, REST APIs

Databases: PostgreSQL, MongoDB, ClickHouse, Redis

Infrastructure & DevOps: Docker, GitHub Actions, AWS (SES), Prisma, Drizzle ORM, BullMQ

Core Concepts: Data Structures & Algorithms, System Design, Distributed Systems, Microservices

Experience

InboxKit

Jan 2026 – Present

Full-Stack Software Engineer

Remote

Tech: Go, Fiber, Fastify, Next.js, ClickHouse, Redis, BullMQ, Puppeteer, AWS SES, Docker, Mongo DB

- **Owned the Enrich Platform end-to-end:** designed and shipped the full product (frontend, backend, infra, deployment), including the REST API surface powering the customer dashboard and an internal admin dashboard for monitoring and abuse detection.
- Designed and built the customer-facing UI in Next.js, Tailwind, and Shadcn: dashboards, billing, analytics, and a reusable component library with accessible, responsive layouts across breakpoints.
- Architected ClickHouse-backed analytics over **300M+ records**, applying **multi-tier Redis caching**, query rewriting, and partitioning to keep dashboard queries at **sub-second p95 latency**.
- Hardened platform **security**: rate limiting, RBAC, signed sessions, strict input validation, and audit logging to prevent abuse, injection, and account takeover.
- Built microservices and automation pipelines (browser automation, transactional email, LinkedIn scraping) with Redis-backed BullMQ queues handling retries, backoff, and rate-limited dispatch.

Key Projects

DevSync – Real-Time Collaborative Coding Platform *Next.js, Prisma, Yjs, Xvfb, WebSocket, Express, Docker* Live Demo

- Built a Next.js collaborative coding platform supporting up to **15 developers editing the same repository simultaneously** in real time via Yjs CRDTs over WebSocket, with low-latency cursor and selection sync.
- Architected a distributed microservices system separating collaboration (WebSocket, CRDT sync, presence) from persistence and execution layers for horizontal scalability.
- Developed a Dockerized execution backend with isolated disk storage, capable of running React, Express, and GUI apps (Tkinter), exposed securely via a reverse proxy.
- Integrated a real-time chat system alongside the editor for seamless team communication during sessions.

GoRedis – Redis-Compatible In-Memory Database (from Scratch) *Go, TCP, RESP Protocol*

GitHub

- Built a Redis-compatible database engine in Go implementing the **RESP wire protocol**, so the official **redis-cli** and client SDKs connect natively, with sub-millisecond GET/SET latency on in-memory operations.
- Implemented core data structures and commands: strings with TTL, lists with blocking ops (**BLPOP**), streams with consumer groups, sorted sets via skip-lists, and geospatial indexing.
- Engineered **MULTI/EXEC transactions** with **WATCH**-based optimistic concurrency control, plus **master-replica replication**, **RDB snapshots**, **AOF write-ahead logging**, Pub/Sub, and auth.

BookWise – Enterprise Library Management System *Next.js, Upstash, Drizzle, Redis, Nodemailer*

Live Demo

- Implemented a protected admin dashboard for user approvals, inventory management, and borrowing workflows.
- Integrated QStash background workflows with Nodemailer to automatically send overdue-book reminder emails.
- Hardened the API surface with rate limiting and request throttling to mitigate abuse and DDoS risk.

Education

Bachelor of Computer Applications (BCA)

2024 – 2027

DAV, Chandigarh, India